Customer Shopping Behavior Analysis

Uncovering insights from 3,900 purchases to drive strategic business decisions

Dataset Overview

3.9K

Purchase Records

Transactional data analyzed

18

Data Fields

Customer and purchase details

37

Missing Values

Review ratings imputed

Analysis Pipeline

Data Preparation

Pandas cleaning, feature engineering, PostgreSQL integration

SQL Analysis

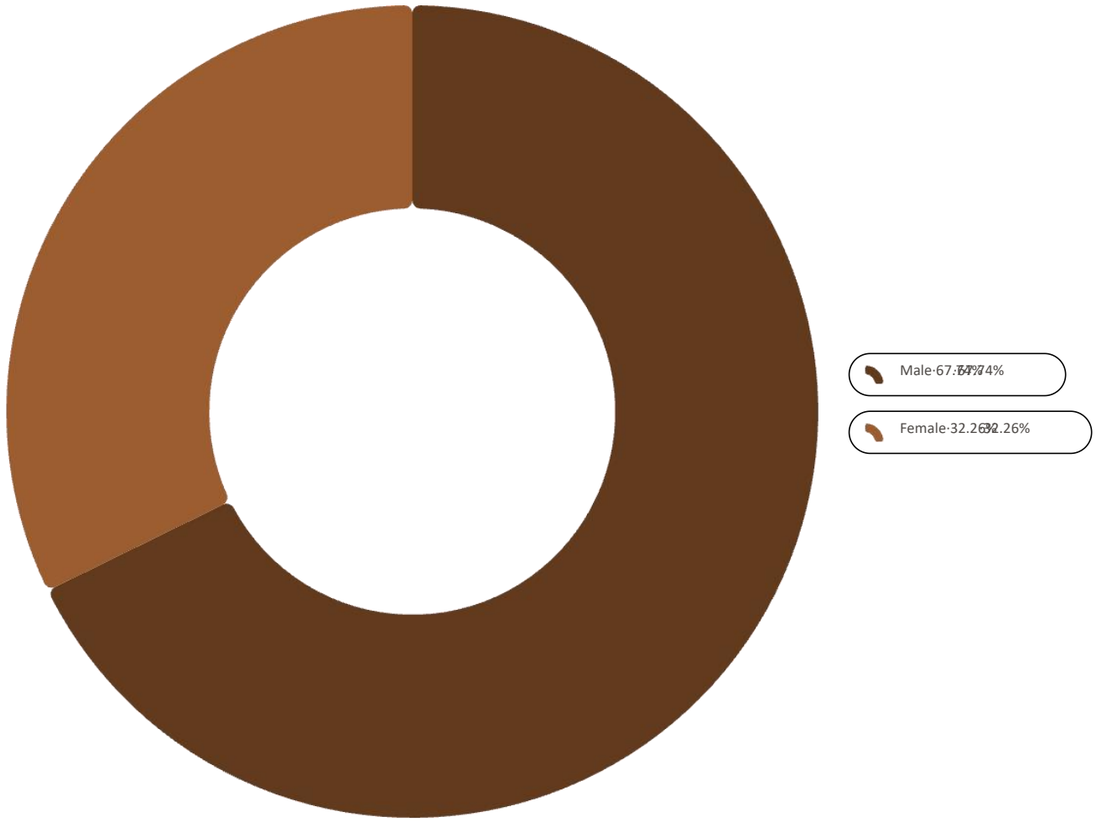
PostgreSQL queries for business insights

Power BI Dashboard

Interactive visualizations and KPIs

Revenue by Gender

Gender, Revenue
Male,2100
Female,1000



Key Finding

Male customers generate 2.1x more revenue than female customers

Product Performance

Top Rated

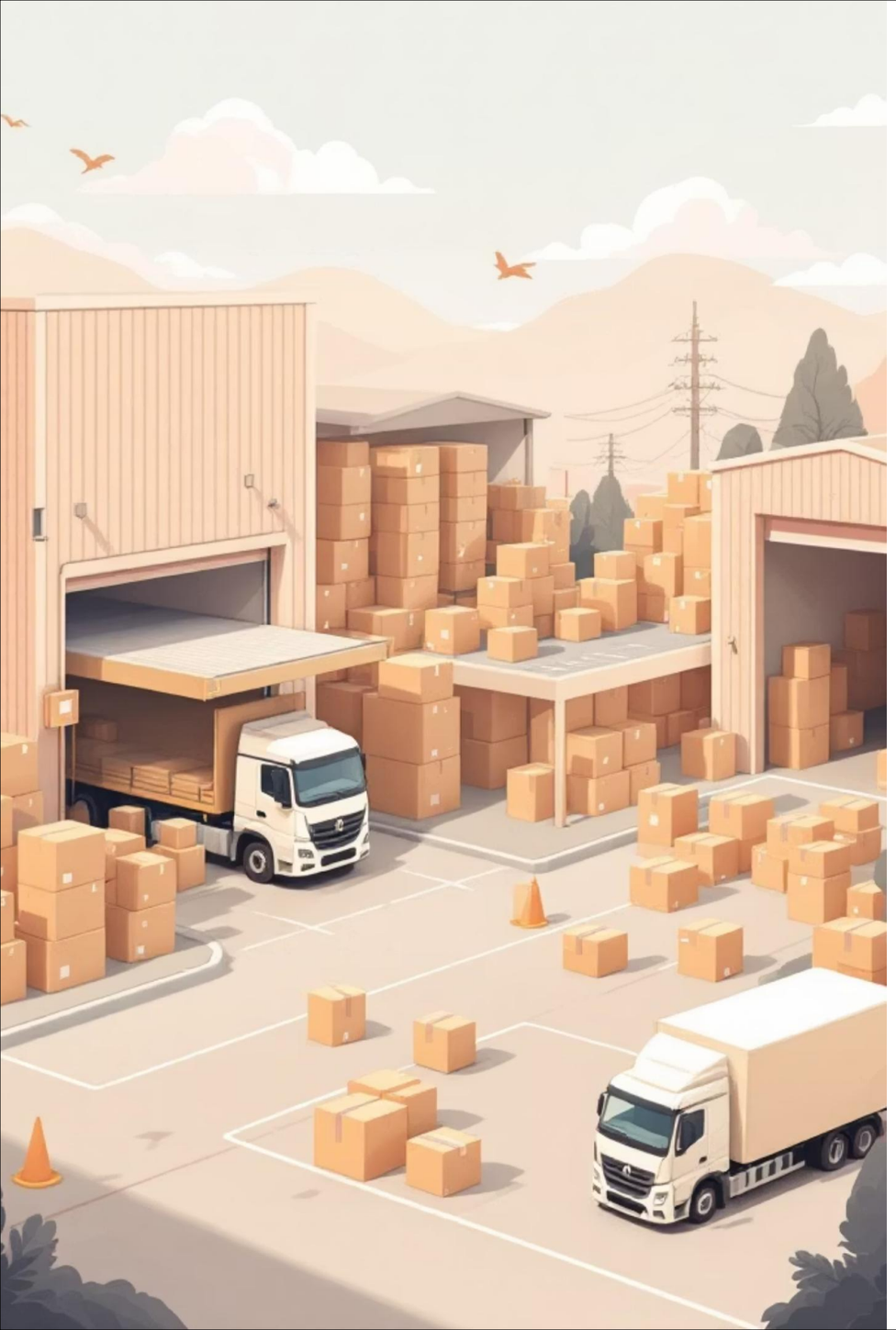
Gloves (3.86), Sandals (3.84), Boots (3.82)

Most Discounted

Hat (50%), Sneakers (49.66%), Coat (49.07%)

Best Sellers

Blouse, Pants, Jewelry (171 orders each)



Shipping & Subscription Impact

Express Shipping

$60.48 average vs. $58.46 standard

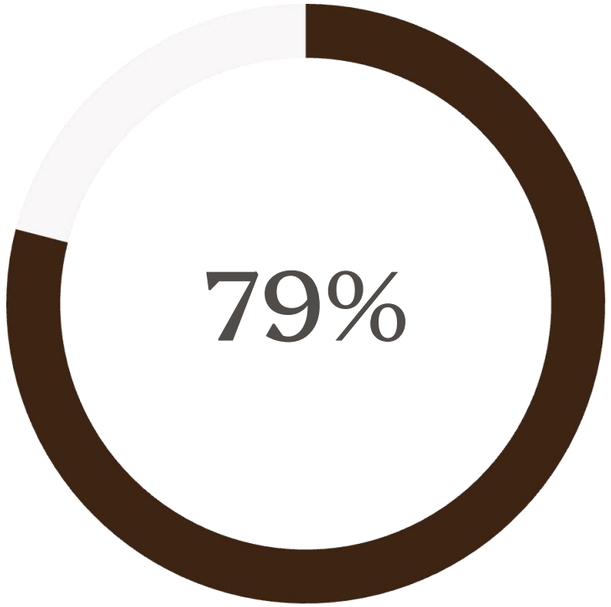
Non-Subscribers

2,847 customers, $170,436 total revenue

Subscribers

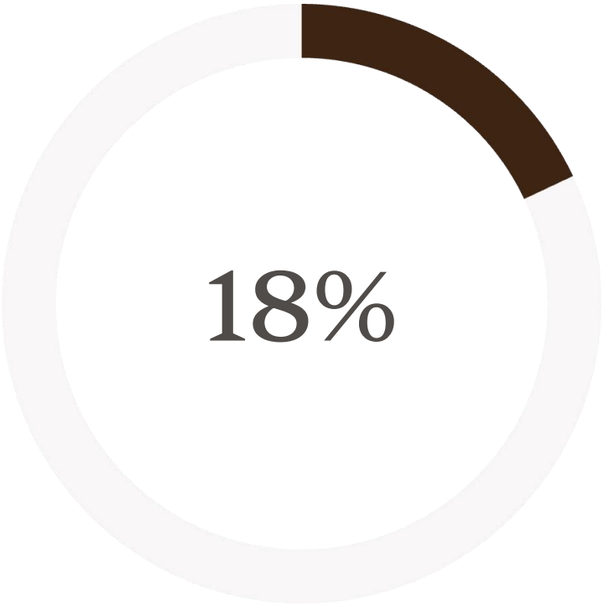
1,053 customers, $62,645 total revenue

Customer Segmentation



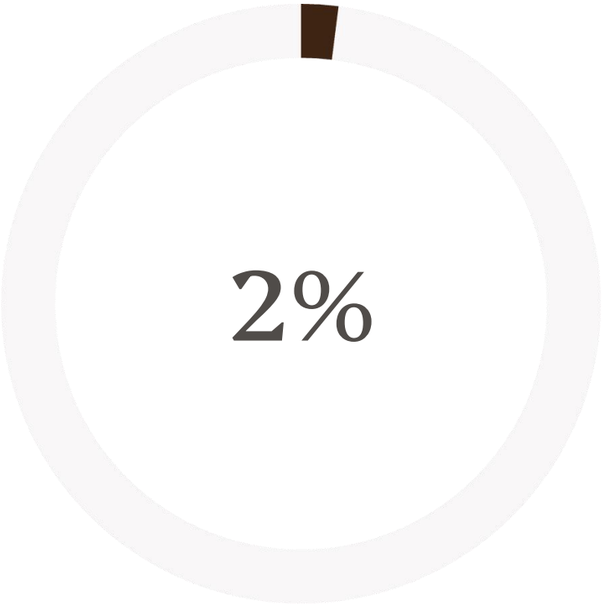
Loyal Customers

3,116 repeat buyers



Returning

701 customers with 2-5 purchases

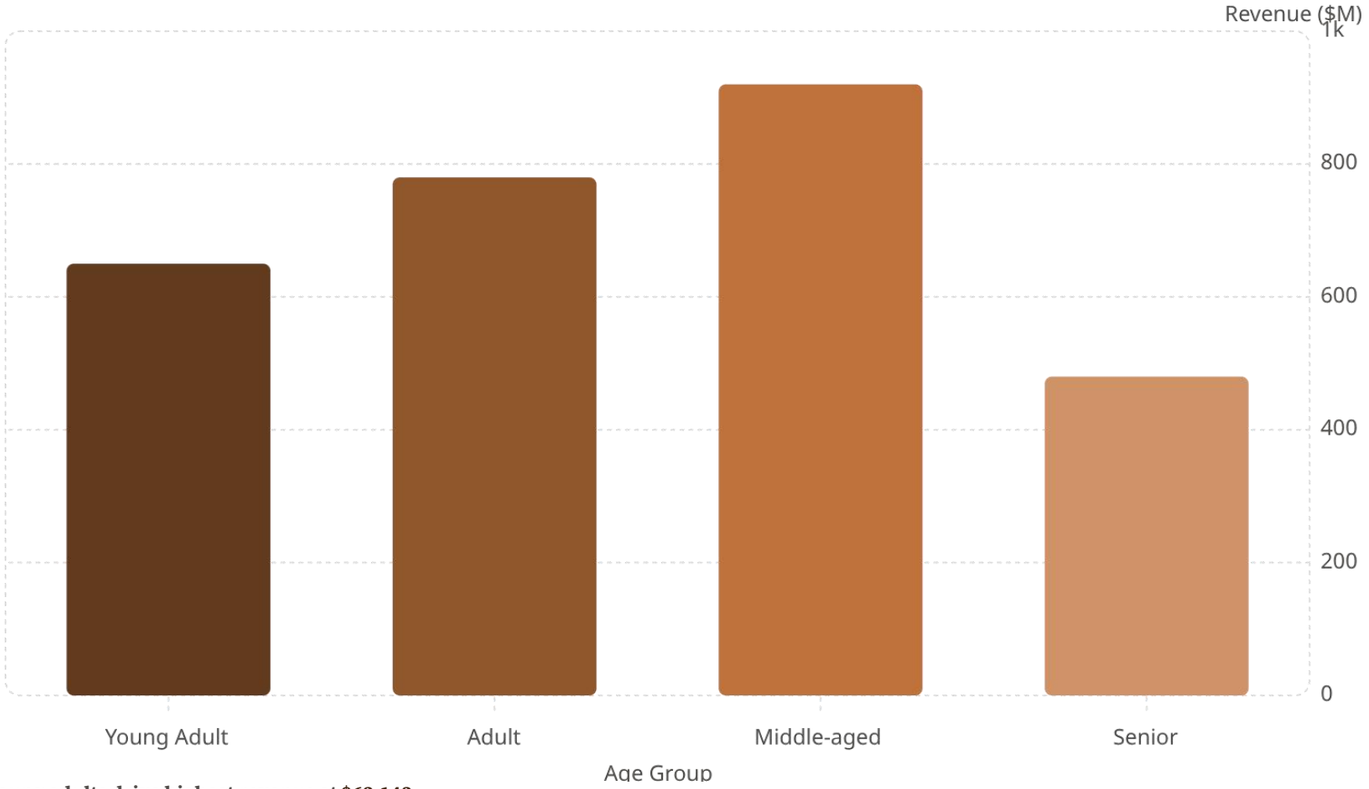


New

83 first-time buyers

Age Group Revenue

Age Group,Revenue
Young Adult,650
Adult,780
Middle-aged,920
Senior,480



Young adults drive highest revenue at $62,143

Key Insights

High-Spending Discount Users

839 customers used discounts while spending above average

Repeat Buyers Subscribe

958 repeat buyers vs. 2,518 non-repeat subscribers

Category Performance

Clothing leads in both revenue and sales volume

Strategic Recommendations

01

Boost Subscriptions

Promote exclusive benefits for subscribers

02

Customer Loyalty Programs

Reward repeat buyers to move them into "Loyal" segment

03

Review Discount Policy

Balance sales boosts with margin control

04

Product Positioning

Highlight top-rated and best-selling products in campaigns

05

Targeted Marketing

Focus on high-revenue age groups and express-shipping users